



CHEMISTRY 4

ACADEMIC

Grade 12

Curriculum Guide

CHEMISTRY 4

Prerequisites: Chemistry 1, Chemistry 2, Chemistry 3

Course Description:

This course introduces the fundamental concepts of organic chemistry, focusing on structure and properties. The learners examine how the unique bonding properties of carbon, together with electronegativity, polarity, and intermolecular forces, determine the structure and behavior of organic molecules. They apply the International Union of Pure and Applied Chemistry (IUPAC) nomenclature to represent organic compounds and analyze hydrocarbons in terms of structure, bonding, and properties, including their importance as fuels and industrial materials.

The course also explores functional groups and major classes of organic compounds, including alkanes, alkenes, alkynes, alkyl halides, alcohols, aromatic compounds, carbonyl compounds, and carboxylic acids. Learners determine the different biomolecules and their importance. Learners analyze how functional groups influence the properties, reactivity, and applications of organic compounds in biological, industrial, and environmental contexts.

Using structural models, comparison of compound classes, and real-world applications, the learners develop a deeper understanding of the relationship between molecular structure and the function of organic compounds and biomolecules.

The course also emphasizes systems thinking and the application of Green Chemistry and the Sustainable Development Goals in analyzing the roles of organic compounds and biochemicals in environmental, health, and industrial contexts.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Foundations of Organic Structure and Hydrocarbons		
1. Introduction to Organic Chemistry	1. carbon has unique bonding properties that enable the formation of a wide variety of organic compounds with diverse structures and functions;	1. describe the unique properties of carbon that lead to the formation of a wide variety of organic compounds;
2. Molecular Structure and Intermolecular Forces	2. bonding, electronegativity, and intermolecular forces determine the structure, polarity, and physical properties of organic compounds;	2. use bonding, electronegativity, and intermolecular forces to explain the properties of organic compounds;
3. Polarity and Electron Distribution	3. electron distribution within molecules influences molecular polarity and stability, which affect physical properties and interactions;	3. use electronegativity differences and dipole moments to explain molecular polarity and properties;
4. Hybrid orbitals and chemical structure	4. how the structures of certain molecules are explained by hybrid orbitals;	4. use hybridization in structure and bonding of organic compounds;
5. Functional Groups	5. functional groups determine the properties and reactivity of organic compounds;	5. identify common functional groups in organic molecules;
6. Alkanes (Saturated Hydrocarbon)	6. hydrocarbons exhibit structural diversity and can be classified based on bonding, which influences their properties and uses; and	6. use IUPAC nomenclature to name and write the structures of alkanes;
		7. analyze structural isomerism in alkanes and relate structure to physical properties;
		8. describe some simple reactions of organic compounds such as combustion, addition, substitution, condensation, and saponification;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
		9. explain the importance of alkanes as fuels and relate their properties to energy use, safety, and global warming;
7. Alkenes and Alkynes (Unsaturated Hydrocarbons)	7. the presence of multiple bonds in hydrocarbons affects structure, reactivity, and applications.	10. compare alkanes, alkenes, and alkynes in terms of structure, bonding, and properties;
		11. use IUPAC nomenclature to name and write the structures of alkenes and alkynes; and
		12. explain the role of alkenes and alkynes as raw materials in the production of polymers and industrial chemicals.
Performance Standards <i>The learners</i> can use structural models to represent organic compounds and write their IUPAC names. They can explain how bonding, polarity, and functional groups determine their physical and chemical properties. They can also analyze and compare hydrocarbons based on structure and bonding and apply these concepts to explain the uses of organic compounds such as fuels and industrial materials in relation to their properties and safety considerations.		
Suggested Performance Tasks - Construct models of organic compounds such as alkanes, alkenes, and alkynes, by showing their structures, IUPAC names, and functional groups, and explain how bonding and polarity influence their properties. - Analyze the structure and properties of selected hydrocarbons (e.g., LPG, gasoline, and acetylene) and explain how their structure and properties influence their uses as fuels or industrial raw materials, including safety and environmental considerations.		
Structure and Applications of Organic Compounds		
8. Alkyl Halides and Alcohols	8. functional groups influence the structure, properties, and uses of organic compounds;	13. use IUPAC nomenclature to name and write the structures of alkyl halides and alcohols;
		14. explain how functional groups influence the properties, reactivities, and uses of alkyl halides and alcohols;
9. Aromatic Compounds	9. aromatic compounds exhibit unique stability due to electron	15. describe the structure and stability of aromatic compounds;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	delocalization, which affects their properties and applications;	16.analyze how aromatic structure influences their properties and uses;
10. Carbonyl Compounds (Aldehydes and Ketones)	10.carbonyl functional groups affect the physical properties and chemical behavior of organic compounds;	17.use IUPAC nomenclature to name and write the structures of aldehydes and ketones;
		18.explain how the carbonyl group influences the properties of aldehydes and ketones;
11. Carboxylic Acids and Derivatives	11.carboxylic functional groups influence acidity, properties, and reactivity of compounds;	19.describe the structure and properties of carboxylic acids and some of their derivatives, such as esters;
12. Biomolecules	12.biomolecules are organic compounds whose structures determine their biological functions; and	20.explain the importance of biomolecules, including carbohydrates, lipids, proteins, and nucleic acids in terms of their structure, function, and role in biological processes;
		21.Conduct experiments showing the properties of biomolecules;
22.Application of Organic Compounds	13. the structure and functional groups of organic compounds determine their applications in various fields.	22. explain how organic compounds are used in industry, such as the use of alkenes in the production of polymers and plastics); and
		23.discuss how polymers have contributed to the development while also causing pollution, health hazards, and other socio-scientific issues.
Performance Standards <i>The learners can use IUPAC nomenclature and structural models to represent organic compounds and explain how functional groups influence their properties and behavior. They can analyze and compare different classes of organic compounds based on their properties, and relate these to their applications in biological, industrial, and environmental contexts. They can also examine the characteristics of biomolecules and discuss their significance in both biological and everyday contexts.</i>		
Suggested Performance Tasks Construct models of different organic compounds, identify their structures and IUPAC names, and explain how their functional groups influence their properties and uses.		

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Analyze a real-world application of organic compounds, such as pharmaceuticals, polymers, or biomolecules and explain how structure and functional groups determine their function and use.		